Machine Learning (ML)

Short theory, keywords and probable viva / exam questions for every practical

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Practical 8: Uber Fare Prediction (Linear Regression and Random Forest)

Topics in short

Regression: Supervised learning that predicts a continuous number (the fare).

Pre-processing: Drop useless columns (IDs), remove missing values, extract date features (hour, weekday, year), compute trip distance with the Haversine formula, remove outliers with the IQR rule (outside Q1 - 1.5*IQR and Q3 + 1.5*IQR).

Linear Regression: Fits y = b0 + b1x1 + ... + bnxn by minimizing squared error (least squares).

Random Forest Regressor: Ensemble of many decision trees trained on bootstrap samples (bagging); output is the average of the trees. Captures non-linear patterns.

Evaluation metrics: R2 = share of variance explained (1 is perfect). MSE = mean squared error. RMSE = sqrt(MSE), in the same unit as the fare. MAE = mean absolute error.

Keywords

regression, supervised learning, feature engineering, Haversine distance, outliers, IQR, box plot, correlation, train-test split, least squares, ensemble, bagging, R2, RMSE, MAE

Probable questions (with short answers)

Q1. Why do we remove outliers?

Ans: Extreme or wrong values (fare <= 0, huge distances) distort the model, especially linear regression.

Q2. What is the IQR rule?

Ans: Values below Q1 - 1.5*IQR or above Q3 + 1.5*IQR are outliers, where IQR = Q3 - Q1.

Q3. What is the Haversine formula?

Ans: It gives the great-circle distance between two latitude/longitude points on a sphere.

Q4. What does R2 mean?

Ans: Fraction of the variance of the target explained by the model; 1 = perfect, 0 = same as predicting the mean.

Q5. RMSE vs MAE?

Ans: RMSE punishes large errors more (squares them); MAE treats all errors equally.

Q6. Why did Random Forest perform better?

Ans: It models non-linear relations and interactions; averaging many trees reduces variance.

Q7. What is correlation?

Ans: A measure (-1 to +1) of how strongly two variables move together linearly.

Q8. Assumptions of linear regression?

Ans: Linearity, independence of errors, constant variance (homoscedasticity), normal errors, no multicollinearity.

Q9. What is overfitting?

Ans: Model learns noise of the training data and performs poorly on new data.

Q10. Why split into train and test sets?

Ans: To measure performance on unseen data.

Practical 9: Email Spam Detection (KNN and SVM)

Topics in short

Binary classification: Predict one of two classes: Not Spam (0, normal) or Spam (1, abnormal).

Dataset: Each row is an email, each column the count of one word (bag-of-words). 3000 features.

K-Nearest Neighbors: Lazy learner: stores training data; for a new email finds the k closest emails (Euclidean distance) and takes a majority vote.

Support Vector Machine: Finds the hyperplane with the maximum margin between the classes. Support vectors are the closest points. Kernels (RBF, linear, polynomial) handle non-linear boundaries; C controls the trade-off between margin and errors.

Metrics: Accuracy, confusion matrix, precision, recall, F1-score.

Keywords

binary classification, bag of words, feature vector, normalization, KNN, lazy learner, Euclidean distance, SVM, hyperplane, margin, support vectors, kernel trick, RBF, confusion matrix, precision, recall, F1-score

Probable questions (with short answers)

Q1. Why is KNN called a lazy learner?

Ans: It does no training; all computation happens at prediction time.

Q2. How do you choose k in KNN?

Ans: Try several values with cross-validation; use an odd k for binary problems to avoid ties.

Q3. What are support vectors?

Ans: Training points closest to the decision boundary; they define the margin.

Q4. What is the kernel trick?

Ans: Computing similarity in a higher-dimensional space without explicitly transforming the data.

Q5. Role of parameter C in SVM?

Ans: Large C = fewer training errors, narrow margin; small C = wider margin, more tolerance.

Q6. Why is SVM better on this data?

Ans: Text data is high-dimensional; SVM handles that well while KNN distances become less meaningful.

Q7. Precision vs recall for spam?

Ans: Precision: of emails marked spam, how many are spam. Recall: of all spam, how many were caught.

Q8. Why normalize the emails?

Ans: Long emails have larger word counts; normalization makes distances comparable.

Q9. What is the curse of dimensionality?

Ans: With many features, data becomes sparse and distances lose meaning, hurting KNN.

Q10. What is F1-score?

Ans: Harmonic mean of precision and recall: 2PR / (P + R).

Practical 10: Bank Customer Churn (Neural Network)

Topics in short

Problem: Predict whether a customer will leave the bank (Exited = 1) in the next 6 months.

Feature and target: Drop IDs and surname; encode Gender (0/1) and Geography (one-hot). Target is Exited.

Normalization: StandardScaler: z = (x - mean) / std. Fit on training data only to avoid data leakage.

Artificial Neural Network: Layers of neurons: input -> hidden (ReLU) -> output (sigmoid gives probability). Trained with backpropagation and the Adam optimizer on binary cross-entropy loss.

Improvements: More neurons, Dropout (regularization), EarlyStopping, class weights for the imbalanced data (only ~20% leave).

Keywords

ANN, perceptron, neuron, weights, bias, activation function, ReLU, sigmoid, forward propagation, backpropagation, gradient descent, Adam, epoch, batch size, loss function, binary cross-entropy, dropout, early stopping, class imbalance, one-hot encoding

Probable questions (with short answers)

Q1. What is an activation function? Why needed?

Ans: Adds non-linearity so the network can learn complex patterns. Examples: ReLU, sigmoid, tanh.

Q2. Why sigmoid in the output layer?

Ans: It squashes output to 0..1, interpreted as the probability of leaving.

Q3. What is backpropagation?

Ans: Computing the gradient of the loss for every weight using the chain rule, from output to input.

Q4. Epoch vs batch size?

Ans: Epoch = one pass over all training data. Batch size = samples per weight update.

Q5. Why normalize the data?

Ans: Features on different scales (Balance vs Age) slow training; scaling helps gradient descent converge.

Q6. What is Dropout?

Ans: Randomly disabling neurons during training to reduce overfitting.

Q7. What is EarlyStopping?

Ans: Stop training when validation loss stops improving.

Q8. Why is accuracy misleading here?

Ans: Data is imbalanced (~80% stay); a model predicting 'stay' always gets ~80%. Look at recall/F1 for 'Left'.

Q9. What is one-hot encoding?

Ans: Converting a category into separate 0/1 columns, one per category.

Q10. What loss function is used and why?

Ans: Binary cross-entropy, the standard loss for two-class probability output.

Practical 11: K-Nearest Neighbors on Diabetes Dataset

Topics in short

Pre-processing: Columns like Glucose, BMI, BloodPressure cannot be 0, so zeros are treated as missing and filled with the median. Outliers checked with the IQR rule. Correlation shows Glucose is the strongest predictor.

KNN: Classifies a patient by majority vote of the k most similar patients. Distance-based, so features must be scaled.

Choosing k: GridSearchCV with 5-fold cross-validation tries many k values and keeps the best.

Metrics from confusion matrix: Accuracy = (TP+TN)/total. Error rate = 1 - accuracy. Precision = TP/(TP+FP). Recall = TP/(TP+FN).

Keywords

KNN, distance metric, Euclidean, Manhattan, feature scaling, missing values, imputation, median, outliers, cross-validation, GridSearchCV, hyperparameter, confusion matrix, TP/TN/FP/FN, accuracy, error rate, precision, recall

Probable questions (with short answers)

Q1. Define TP, TN, FP, FN.

Ans: TP: diabetic predicted diabetic. TN: healthy predicted healthy. FP: healthy predicted diabetic. FN: diabetic predicted healthy.

Q2. Formula for error rate?

Ans: Error rate = (FP + FN) / total = 1 - accuracy.

Q3. Which matters more here, precision or recall?

Ans: Recall: missing a diabetic patient (FN) is more dangerous.

Q4. Why scale features for KNN?

Ans: Distance is dominated by large-valued columns (Insulin) otherwise.

Q5. Effect of small and large k?

Ans: Small k: noisy, overfits. Large k: smooth, may underfit.

Q6. What is cross-validation?

Ans: Split the training data into k folds; train on k-1 folds and test on the remaining one, repeated k times.

Q7. Why fill zeros with the median instead of the mean?

Ans: The median is not affected by outliers.

Q8. What is data leakage?

Ans: Using test data information (e.g. its statistics) during training, which gives over-optimistic scores.

Q9. Distance measures used in KNN?

Ans: Euclidean, Manhattan, Minkowski, cosine.

Q10. Is KNN parametric?

Ans: No, it is non-parametric and instance-based.

Practical 12: K-Means and Hierarchical Clustering (Mall Customers)

Topics in short

Unsupervised learning: No labels; the algorithm finds groups (clusters) in the data. Used for customer segmentation.

K-Means: 1) pick k centroids (k-means++), 2) assign each point to the nearest centroid, 3) move each centroid to the mean of its points, 4) repeat until no change.

Elbow method: Plot WCSS (within-cluster sum of squares) for k = 1..10; the bend ('elbow') where the drop slows down is the best k (here k = 5).

Hierarchical clustering: Agglomerative (bottom-up): start with each point as a cluster and merge the closest pair repeatedly. The dendrogram shows the merges; cutting it gives clusters. Ward linkage merges the pair with the smallest increase in variance.

Silhouette score: From -1 to 1; higher means points are close to their own cluster and far from others.

Keywords

unsupervised learning, clustering, centroid, K-Means, k-means++, WCSS, inertia, elbow method, silhouette score, hierarchical clustering, agglomerative, divisive, dendrogram, linkage (single, complete, average, Ward), customer segmentation

Probable questions (with short answers)

Q1. Supervised vs unsupervised learning?

Ans: Supervised uses labelled data (classification, regression); unsupervised finds structure without labels.

Q2. What is WCSS?

Ans: Sum of squared distances of points from their cluster centroid.

Q3. How does the elbow method work?

Ans: Choose k where adding more clusters gives only a small decrease in WCSS.

Q4. What is k-means++?

Ans: Smart initialization that places initial centroids far apart, giving better and faster results.

Q5. Limitations of K-Means?

Ans: Need to choose k, sensitive to outliers and initial centroids, assumes round clusters of similar size.

Q6. Agglomerative vs divisive?

Ans: Agglomerative: bottom-up merging. Divisive: top-down splitting.

Q7. What is a dendrogram?

Ans: Tree diagram showing the order and distance at which clusters merge.

Q8. Types of linkage?

Ans: Single (min distance), complete (max), average, Ward (minimum variance).

Q9. What does the silhouette score tell us?

Ans: How well-separated the clusters are; near 1 = good, near 0 = overlapping.

Q10. Business use of the 5 clusters?

Ans: Target high-income high-spending customers with premium offers, low spenders with discounts, etc.